MTH 451/551
Lecture 5

Prof. W. Pazner

1 Solving Equations and Root Finding

1.1 Newton and Secant Methods

[Textbook reference: §2.2]

Recall that the regula falsi method used the root of the secant line to find the next approximation to the root of $f(x)$. The idea is that the secant line can be thought of as an approximation to $f(x)$, and its zero is easily computable. This same idea of **linear approximation** underlies both of the methods that we will see in this section: Newton's method and the secant method.

These methods are not root bracketing methods. They do not require that we start with an interval that contains the root, but they also do not guarantee that we obtain nested intervals containing the root. In fact, these methods may sometimes *fail to converge*. However, when they do converge, they usually do so *much more rapidly* than the methods we have considered earlier. This can be seen as a trade-off between robustness and speed of convergence. We briefly note that there are ways to combine these fast methods with robust root bracketing methods to obtain methods that are both fast and robust. The combination of the secant method with bisection is called "Brent's method", and the combination of Newton's method with bisection is often called "Newt-safe".

1.1.1 Newton's Method

Newton's method starts with an "initial guess" x_0, which should be "close" to a root r of $f(x)$, and then proceeds by approximating $f(x)$ by its **linearization**, i.e. first-order Taylor polynomial taken at the point x_0,

$$p_1(x) = f(x_0) + f'(x_0)(x - x_0).$$

Then, the root of $f(x)$ is approximated by the root of $p_1(x)$,

$$f'(x_0)x = f'(x_0)x_0 - f(x_0) \qquad \Longrightarrow \qquad x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

We call this new point x_1. The hope is that x_1 is a better approximation than x_0 (i.e. closer to the true root r). Then, if we repeat this process, starting now at x_1, we will get an even better approximation x_2, and so on. This gives the iteration:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

(as long as $f'(x_n) \neq 0$).

The secant method is very similar. Instead of approximating $f(x)$ by its first-order Taylor polynomial, we approximate it by its secant line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$. This requires two initial guesses, $x_0 \neq x_1$. Then, the root of the secant line is given by

$$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1) = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}.$$

Then, we proceed by using the most recent two points for the secant line, obtaining the iteration

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n) = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}.$$

The secant method requires evaluation only of the function f, whereas Newton's method requires evaluation of the function f and its derivative f'.

We summarize Newton's method and the secant method in the algorithms below.

```
1:  function NEWTON(f, f′, x, ftol, xtol, maxit)
2:      for n = 1, 2, ..., maxit do
3:          δ ← f(x)/f′(x)
4:          x ← x − δ
5:          if |f(x)| ≤ ftol or |δ| ≤ xtol then
6:              return x
7:          end if
8:      end for
9:      Did not converge.
10: end function
```

```
1:  function SECANT(f, x0, x1, ftol, xtol, maxit)
2:      for n = 1, 2, ..., maxit do
3:          δ ← (x1 − x0)f(x1)/(f(x1) − f(x0))
4:          x0 ← x1,  x1 ← x1 − δ
5:          if |f(x1)| ≤ ftol or |δ| ≤ xtol then
6:              return x1
7:          end if
8:      end for
9:      Did not converge.
10: end function
```

1.1.2 Differences Between Regula Falsi and Secant Method

Both the regula falsi and the secant methods use the root of the secant line as the next point in the iteration, and so they appear to be very similar. However, there is a key difference: the secant method **always uses the newest two points** to form the secant line. Regula falsi, on the other hand, uses the secant line between the **newest point** and **one of the previous points**, chosen so that $f(x)$ has opposite sign at the two endpoints (ensuring that the root is bracketed). Other than this difference, the methods are the same. Although this difference appears minor, it can have significant implications for robustness and speed of convergence. Regula falsi guarantees that

the root remains bracketed, and is guaranteed to converge as long the starting interval is valid, but convergence may be slow. Heuristically, secant method should converge faster because it is always using the most up-to-date information: we expect $x_n \to r$, so the most recent points x_n and x_{n-1} should be the best approximations so far to the root r, rather than using a_n or b_n as one of the endpoints, which may be "stale" information. The secant method does indeed converge (much) faster than regula falsi, and we will make this argument rigorous shortly. The drawback is that it is not guaranteed to converge even if the initial two points bracket the root. Additional criteria are needed to guarantee convergence, and these generally depend on unknown parameters that cannot be checked practically.

Example 1. As an example, we illustrate the use of the regula falsi and secant methods to solve

$$x^2 - 2 = 0.$$

The roots are $\pm\sqrt{2} = 1.41421\ldots$, and so we use the interval $[1, 2]$ to bracket the root (and for the secant method initialization). The results are summarized in the table below, with the correct digits of the approximations $x_n \approx \sqrt{2}$ shown in bold.

	Regula Falsi			Secant	
n	a_n	x_n	b_n	n	x_n
0	1.000000000000000	**1**.333333333333333	2.000000000000000	0	1.000000000000000
1	1.333333333333333	**1.4**00000000000000	2.000000000000000	1	2.000000000000000
2	1.400000000000000	**1.41**1764705882353	2.000000000000000	2	**1**.333333333333333
3	1.411764705882353	**1.41**3793103448276	2.000000000000000	3	**1.4**00000000000000
4	1.413793103448276	**1.414**141414141414	2.000000000000000	4	**1.414**634146341463
5	1.414141414141414	**1.4142**01183431953	2.000000000000000	5	**1.41421**1438474870
6	1.414201183431953	**1.41421**1438474870	2.000000000000000	6	**1.4142135620**57320
7	1.414211438474870	**1.414213**197969543	2.000000000000000	7	**1.414213562373095**
8	1.414213197969543	**1.414213**499851323	2.000000000000000	8	**1.414213562373095**
9	1.414213499851323	**1.4142135**51646055	2.000000000000000	9	**1.414213562373095**
10	1.414213551646055	**1.4142135620**532626	2.000000000000000	10	**1.414213562373095**
				11	**1.414213562373095**
				12	**1.414213562373095**

The secant method gets all digits correct in 5 iterations, whereas the regula falsi method, even after 10 iterations, does not have all digits correct. ♦

Theorem 1 (Convergence of Newton's method; Textbook Theorem 2.12)**.** *Suppose that $r \in (a, b)$ is a root of $f \in C^2(a, b)$, and $f'(r) \neq 0$. There is a $\delta > 0$ such that, if $x_0 \in (r - \delta, r + \delta)$, then the Newton iterates $\{x_n\}$ will be well-defined and in $(r - \delta, r + \delta)$ for $n \geq 0$, and $x_n \to r$ at least quadratically, with*

$$\frac{x_{n+1} - r}{(x_n - r)^2} \to \frac{f''(r)}{2f'(r)} \qquad \text{as } n \to \infty,$$

provided $x_n \neq r$ for any n. If $f''(r) \neq 0$, the convergence of Newton's method is precisely quadratic.

Proof. Let $x_n \in (a,b)$ with $f'(x_n) \neq 0$. Let p be the tangent line through $(x_n, f(x_n))$, i.e. the first-order Taylor polynomial of f at x_n,

$$p(x) = f(x_n) + f'(x_n)(x - x_n).$$

Taylor's Theorem gives

$$f(x) = p(x) + \frac{f''(\xi)}{2}(x - x_n)^2$$

for some ξ between x_n and x. Evaluating at the root r then gives

$$-p(r) = \frac{f''(\xi_n)}{2}(r - x_n)^2$$

where ξ_n is between x_n and r. The next point x_{n+1} in Newton's method is the root of p, i.e. $p(x_{n+1}) = 0$. So,

$$\begin{aligned} -p(r) &= p(x_{n+1}) - p(r) \\ &= f(x_n) + f'(x_n)(x_{n+1} - x_n) - f(x_n) - f'(x_n)(r - x_n) \\ &= f'(x_n)(x_{n+1} - r). \end{aligned}$$

Equating these two gives

$$x_{n+1} - r = \frac{f''(\xi_n)}{2f'(x_n)}(x_n - r)^2. \tag{1}$$

We claim that the ratio $\frac{f''(\xi_n)}{2f'(x_n)}$ is bounded as long as x_n is sufficiently close to the root r. To see this, note that since $f \in C^2(a,b)$, $f''(x)$ is bounded above close to r. Additionally, since f' is continuous in (a,b) and $f'(r) \neq 0$, $f'(x)$ is bounded below close to r. Therefore, there exists some $\delta > 0$ such that

$$C \coloneqq \frac{\max\limits_{x \in [r-\delta, r+\delta]} |f''(x)|}{2 \min\limits_{x \in [r-\delta, r+\delta]} |f'(x)|} < \infty.$$

We can choose δ even smaller to ensure $\gamma \coloneqq C\delta < 1$. Then, if $x_n \in (r-\delta, r+\delta)$, (1) gives

$$|x_{n+1} - r| \leq C|x_n - r|^2 \leq \gamma |x_n - r| < |x_n - r| < \delta, \tag{2}$$

so $x_{n+1} \in (r-\delta, r+\delta)$ as well. This means that if $x_0 \in (r-\delta, r+\delta)$, then x_n will be for $n = 1, 2, \ldots$. Let $e_n = x_n - r$. From (2),

$$|e_n| \leq \gamma^n |e_0| \gamma^n (b-a).$$

Since $\gamma < 1$, this implies $|e_n| \to 0$, and so $x_n \to r$. The remainder of the theorem statements follow from (1). □

We have a similar result for the secant method.

Theorem 2 (Convergence of the secant method; Textbook Theorem 2.13)**.** *Suppose that $r \in (a, b)$ is a root of $f \in C^2(a, b)$, and $f'(r) \neq 0$. There is a $\delta > 0$ such that, if $x_0, x_1 \in (r - \delta, r + \delta)$, then the secant iterates $\{x_n\}$ will be well-defined and in $(r - \delta, r + \delta)$ for $n \geq 0$, and $x_n \to r$ with at least order $\alpha = (1 + \sqrt{5})/2 \approx 1.61803$ (the golden ratio), with*

$$\frac{|x_{n+1} - r|}{|x_n - r|^\alpha} \to \left| \frac{f''(r)}{2f'(r)} \right|^{\alpha - 1} \qquad \text{as } n \to \infty,$$

provided $x_n \neq r$ for any n. If $f''(r) \neq 0$, the convergence is precisely of order α.

The proof is given in Exercises 15 and 16 of §2.2 of the textbook.